

LiteLLM

1. What is LiteLLM?

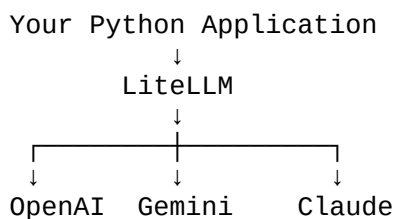
LiteLLM is a Python library that provides a common interface for communicating with different Large Language Models (LLMs).

It allows your application to use models from different providers such as:

- OpenAI
- Google Gemini
- Anthropic Claude
- Azure OpenAI
- Local models, etc.

Instead of learning a different API format for every provider, you can use a similar LiteLLM interface.

Simple idea



2. Is LiteLLM itself an LLM?

No.

LiteLLM is **not a model**.

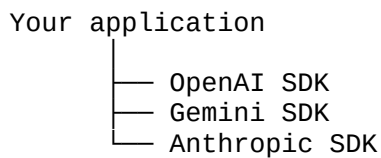
Component	What it is
GPT	LLM
Gemini	LLM
Claude	LLM
LiteLLM	Library/interface for communicating with LLMs

The actual response is generated by the LLM, not by LiteLLM.

3. Why do we use LiteLLM?

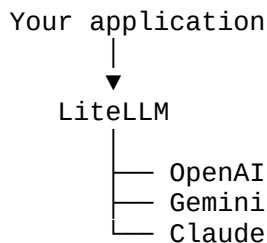
Different LLM providers have different APIs.

Without LiteLLM:



You may need provider-specific code.

With LiteLLM:



So LiteLLM provides a **common interface** and makes it easier to switch between providers.

4. Does LiteLLM provide API keys?

No.

Installing LiteLLM:

```
pip install litellm
```

does **not** give you access to GPT, Gemini, Claude, etc.

You need credentials from the provider whose model you want to use.

For example:

```
OpenAI model → OpenAI API key
Gemini model → Google API key
Claude model → Anthropic API key
```

5. How are API keys handled?

The recommended approach is to store the API key in an **environment variable**, rather than writing it directly in your Python code.

For OpenAI:

```
OPENAI_API_KEY=your_api_key
```

Then Python/LiteLLM can access it through the environment.

Don't do this:

```
completion(
    model="openai/gpt-4",
```

```
)    api_key="your_actual_key"
```

Especially if the project is going to GitHub.

Prefer:

```
.env
↓
Environment variable
↓
LiteLLM
↓
OpenAI API
```

And add .env to .gitignore.

```
.env
venv/
```

I.e The usage steps are :

1. Get an api key of your model from the corresponding provider.
2. Create a file '.env' in your project directory and keep the api key in it .like:
 OPENAI_API_KEY=your_api_key

This is done inorder to keep the api key safe. Giving the api key directly in a python code is unsafe and unprofessional.

3. Now to access this api key from the '.env' file , we need install 'python-dotenv'
Inside your virtual environment:

```
via
pip install python-dotenv
```

4. Then add the following codes on top of your python code :

```
from litellm import completion
from dotenv import load_dotenv
load_dotenv()
```

5. Now your litellm can access the models via these api keys and give you response from the models.

6. Dont forget to add .env to the .gitignore file else it will be added to the github when we perform git push, which in turn could go public.
I.e in .gitignore:

```
.env
venv/
__pycache__/
```

